

DevSage AI – AI Codebase Assistant

Product Requirements Document (PRD)

1. Project Overview

DevSage AI is an AI-powered assistant that understands an entire GitHub repository and answers project-specific questions using Retrieval-Augmented Generation (RAG). It helps developers onboard faster, explore codebases, and generate documentation.

2. Problem Statement

Developers spend significant time understanding unfamiliar codebases. Existing AI assistants lack repository-specific context, leading to inaccurate answers.

3. Objectives

- Reduce developer onboarding time
- Enable AI-powered code exploration
- Generate trustworthy answers with source citations
- Visualize project architecture
- Improve developer productivity

4. Target Users

Software Developers, Tech Leads, Engineering Managers, Open Source Contributors, and Students.

5. Core Features

- GitHub Repository Import
- AI Chat over Codebase
- RAG Pipeline
- Semantic Code Search
- Architecture Visualization
- API Explorer
- Dependency Analysis
- AI Documentation Generator
- Pull Request Review
- Interview Preparation Mode

6. AI Architecture

Repository → Parser → Code Chunking → Embeddings → Vector Database (ChromaDB/Pinecone) → LangGraph Agents → LLM (OpenAI/Gemini) → React Dashboard

7. Tech Stack

Frontend: React.js, TypeScript, Tailwind CSS, React Flow, Monaco Editor

Backend: FastAPI, Python, LangChain, LangGraph, Celery

AI: OpenAI/Gemini, ChromaDB/Pinecone, Sentence Transformers

Database: PostgreSQL, Redis

DevOps: Docker, GitHub Actions, Vercel, Render/Railway

8. Functional Requirements

- Login with GitHub
- Import repositories
- Ask natural language questions
- View architecture diagrams
- Generate documentation

- Review pull requests
- Export reports

9. Non-Functional Requirements

- Chat response under 3 seconds
- Repository indexing under 60 seconds
- Secure authentication and API key management
- Scalable asynchronous processing
- 99.9% availability target

10. Success Metrics

- >90% retrieval accuracy
- <5% hallucination rate
- >4.5/5 user satisfaction
- <3 sec AI response time

11. Future Enhancements

VS Code Extension, Jira Integration, Slack Integration, Multi-repository Search, Voice Assistant, AI Pair Programmer.